

Fullstack Internship

Technical test



Version	Date	Revised by
1.0	January 2025	josef@hilight.aero
1.1	May 2026	Jules Sanglier

This technical test is designed to assess the three important aspects for doing an internship at HyLight:

- Demonstrating expertise
- Being autonomous, stepping out of your comfort zone and starting from scratch
- A smart conception, design and architecture is more important than tech and tools

Exercise - Display tagged pictures on a map

Estimated time: 4h

The goal of the test is to build a simple web application from scratch. **The app should allow users to sign up and log in, upload geotagged photos, and display them on an interactive map.** When a user clicks on a photo marker, they should be able to view the image and add comments related to what is shown in it. *Integrating an AI feature that automatically generates a description of the picture would be highly appreciated.*

The Strategist

Before writing any code, describe how you would build this application in a clean, scalable, and production-ready way. This description should include:

- The technologies and tools you would choose for the frontend, backend, database, storage, authentication, map display, and AI.
- How you would structure the system (architecture diagram or clear explanation of services, APIs, authentication flow, database models, storage logic, etc.).
- How data flows from image upload to display on the map.
- How you would deploy the application (locally or cloud).
- Describe the trade-offs considered for storing the images and how you would ensure that the map remains performant when visualizing a large dataset of markets (e.g, handling 10k photos).
- A realistic estimation of the time required for each step (setup, backend, frontend, authentication, map integration, testing, and deployment).
This part should be written as if you were preparing the project plan for real production work.

The Builder

Build a functional version of the application. It doesn't need to be perfect or fully optimized, it just needs to work end to end: login, upload geotagged images, display them on a map, and allow comments. *The AI part is optional but very appreciated.*

HyLight Fullstack - Technical test

The final delivery can include a GitHub repository with instructions. A short video demo is optional but appreciated.

Geotagged pictures assets :  Fullstack pictures